

Week#01

PROJECT 1: LINUX & COMMAND LINE BASICS

Author: DevOps Intern

Domain: DevOps Engineering

Batch: 2026

Status: Completed & Verified

1. Executive Summar

This project demonstrates basic command-line system administration and server setup in a Linux environment. All the required tasks were performed using terminal commands without relying on a Graphical User Interface (GUI).

The project included creating directories, creating configuration files, adding log entries, inspecting the directory structure, renaming a log file for backup purposes, checking file permissions and ownership, and verifying the final directory state.

2. Workflow Verification & Evidence

Step 1: Directory Scaffolding

Created the required application and logging directories using the recursive `-p` option.

- **Command:**

```
sudo mkdir -p /app/logs
```

The command created the `/app` directory along with the `/app/logs` subdirectory.

Step 2: Configuration Creation

Created an empty configuration file inside the `/app` directory.

- **Command:**

```
sudo touch /app/config.conf
```

This created the `config.conf` file, which was used as the configuration placeholder for the project.

Step 3: Log Entry Injection

Added the initial status message to the server log file using `echo` and `tee`.

- **Command:**

```
echo "Started" | sudo tee /app/logs/server.log
```

The command created the `server.log` file inside the logs directory and added the `Started` message to it.

Step 4: System Hierarchy Audit

Verified the current working directory and inspected the complete `/app` directory structure recursively.

Command:

```
pwd && sudo ls -R /app
```

This command was used to verify the location and confirm that the required directories and files had been created correctly.

Step 5: Log File Archival

Renamed the active log file to create a backup version of the log.

Command:

```
sudo mv /app/logs/server.log /app/logs/server.bak
```

The `server.log` file was renamed to `server.bak`, demonstrating a basic log backup or rotation operation.

Step 6: Security Audit

Inspected the configuration file to verify its permissions, owner, group, and file information.

Command:

```
sudo ls -l /app/config.conf
```

This command displayed the file's permission settings along with its owner and group information.

Step 7: Final Directory State

Performed a final recursive listing of the `/app` directory to verify the completed project structure.

- **Command:**

```
sudo ls -R /app
```

The final listing confirmed that the required directory structure and files were present, including the renamed backup log file.

3. Technical Learnings & Takeaways

- **mkdir -p:** Used to create complete directory paths in a single command without manually creating each parent directory.
- **touch:** Used to create an empty configuration file.
- **Piping (|) and tee:** Used to send command output to a file while working with elevated administrative privileges.
- **mv:** Used to rename the log file from `server.log` to `server.bak`, demonstrating a basic file backup or rotation operation.
- **ls -l:** Used to inspect file permissions, ownership, group information, and file metadata.
- **ls -R:** Used to inspect the directory structure recursively and verify the final state of the project.

4. Conclusion

This project provided practical experience with Linux command-line operations and basic system administration. I learned how to create directories and files, add information to log files, inspect directory structures, rename files for backup purposes, and check file permissions and ownership.

The project also helped me become more comfortable with working through the Linux terminal and understanding common commands used in a DevOps environment.